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N+ NETWORK FUNDAMENTALS AND BUILDING NETWORKS

DAY:2

TASKS:1

Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

RESULT

1. Opened Wireshark on my laptop.
2. Selected the active Wi-Fi network interface.
3. Clicked Start Capture.
4. Opened www.flipkart.com in my browser and refreshed the homepage.
5. Returned to Wireshark and clicked Stop Capture.
6. Examined the captured packets and identified protocols from different TCP/IP layers.

TCP/IP Layer	Protocol	Reason
Application Layer	HTTPS	Used for secure communication with Flipkart website.
Transport Layer	TCP	Responsible for reliable delivery of data packets.

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Internet Layer	IP	Contains source and destination IP addresses.
Network Access Layer	Ethernet II	Transfers data over the local network using MAC addresses.

TASKS:2

Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

RESULT

OSI Model	TCP/IP Model	Real-World Example
Application	Application	Sending a WhatsApp message
Presentation	Application	Instagram images and HTTPS encryption
Session	Application	Logging into Instagram
Transport	Transport	TCP delivering WhatsApp messages
Network	Internet	Router forwarding IP packets
Data Link	Network Access	Switch sending frames using MAC addresses
Physical	Network Access	Wi-Fi signals or Ethernet cable

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TASKS:3

Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines. Look for fields like Source Port, Destination Port, Sequence Number, etc.

RESULT

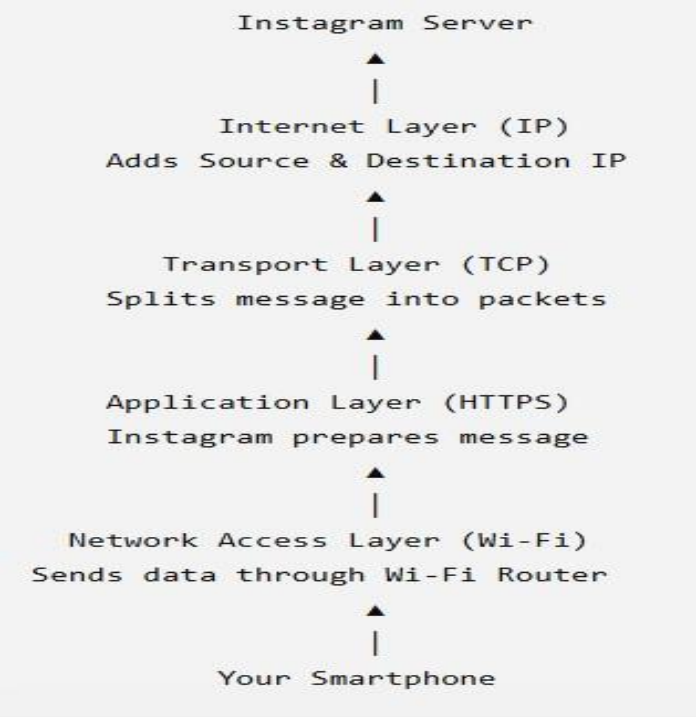
Field	Function
Source Port	Identifies the sending application.
Destination Port	Identifies the receiving application.
Sequence Number	Keeps packets in the correct order.
Acknowledgment Number	Confirms that data has been received successfully.
Window Size	Controls the amount of data sent before receiving acknowledgment.
Checksum	Detects errors in transmitted data.
Flags (SYN, ACK, FIN)	Used to start, maintain, and end TCP connections.

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TASKS:4

Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

RESULT



TASKS:5

Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in

an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

RESULT

I selected an HTTPS packet from my Wireshark capture.

- TCP/IP Layer: Application Layer
- How I Identified It: The Protocol column showed HTTPS, indicating secure communication between my browser and the Flipkart server.
- Real-World Example: This packet represents loading the Flipkart homepage or placing an order securely through the website.